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MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

M-302

MATHEMATICS-III

Time Allotted: 3 Hours | Full Marks: 70

The questions are of equal value.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable.
All symbols are of usual significance.

GROUP A
(Short Answer Type Questions)

1. Answer any *ten* questions. 10×2 = 20

(i) Obtain the period of $\cos 5x$.

(ii) If $f(x) = |x|$, $-\pi < x \le \pi$, then find the Fourier coefficient b_n.

(iii) If $f(x) = 1$, $|x| \le 1$, and $= 0$, $|x| > 1$, then find Fourier Transform of $f(x)$.

(iv) If $f(x) = 1$, $0 \le x \le \pi$, and $= 0$, $x > \pi$, then find Fourier Sine Transform of $f(x)$.

(v) Prove that $f(z) = \bar{z}$ is everywhere continuous but not differentiable.

(vi) Prove that $f(z) = \begin{cases} \dfrac{xy}{x^2+y^2}, & z \neq 0 \\ 0, & z = 0 \end{cases}$

is discontinuous at $z = 0$.

(vii) Locate and name the singularities in the finite z-plane for the $f(z) = \dfrac{z}{(z^2+4)^2}$.

(viii) Evaluate $\int_C \dfrac{zdz}{z-1}$, where $C\,|z| = \dfrac{1}{2}$.

(ix) Show that the probability that exactly one of the events A and B occur is $P(A) + P(B) - 2P(AB)$.

(x) What is the chance that a leap year selected at random will contain 53 Wednesday?

(xi) A random variable X has the density function $f(x) = x$, $0 < x < 1$, $f(x) = \dfrac{1}{2}$, $1 < x < 2$. Find the mean of X.

(xii) The probability density function of a random variable X is $f(x) = k(x-1)(2-x)$, for $1 < x < 2$. Determine the value of the constant k.

(xiii) Express the polynomial $2x^2 + x + 3$ in terms of Legendre's polynomials.

(xiv) Define Bessel's function of 1st kind of order n, J_n.

(xv) Form the PDE by eliminating the arbitrary constants from $az + b = a^2x + y$.

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GROUP B
(Long Answer Type Questions)

Answer any *five* questions taking at least one question from each Module. $5 \times 10 = 50$

Module-I

2. (a) Find a series of cosine multiples of x which will represent $f(x) = x$, $0 \le x \le 2$ and write parseval's identity corresponding to it. Hence, show that $\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \ldots\ldots = \frac{\pi^4}{90}$. 3+2+1

   (b) Find the inverse Fourier Transform of the function $\frac{1}{s^2 + 6s + 25}$. 4

3. (a) Find the Fourier transform of $xe^{-\frac{25x^2}{2}}$. 4

   (b) Find the Fourier series of $f(x) = \begin{cases} 0, & -\pi < x < a \\ 1, & a \le x \le b \\ 0, & b < x \le \pi \end{cases}$ 6

   and prove that $\sum_{n=1}^{\infty} \frac{\sin n(b-a)}{n} = \frac{\pi - b + a}{2}$

Module-II

4. (a) If $f(z) = \frac{xy^2(x+iy)}{x^2 + y^4}$, $z \ne 0$ 5
   $= 0$, $z = 0$.
   Then prove that $\frac{f(z) - f(0)}{z} \to 0$ as $z \to 0$ along any radius vector but not as $z \to 0$ in any manner.

   (b) If $u = \frac{1}{2}\log(x^2 + y^2)$ find the analytic function $f(z) = u + iv$. 5

5. (a) Prove that $\frac{1}{2\pi i}\int_C \frac{e^{zt}}{z^2 + 1} dz = \sin t$, where $t > 0$ and C is the circle $|z| = \pi$. 5

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   (b) Expand $f(z) = \frac{1}{(z+1)(z+3)}$ in Laurent's series valid for the region $0 < |z+1| < 2$. 5

Module-III

6. (a) The probability that a doctor diagnoses a patient correctly is 80%. The probability that a patient dies after correct diagnosis is 40% and the probability that he dies after wrong diagnosis is 70%. The patient died. What is the probability that the disease was correctly diagnosed? 5

   (b) The radius of a circle has distribution given by the p.d.f. 5
   $$f(x) = \begin{cases} 1, & 1 < x < 2 \\ 0, & \text{otherwise} \end{cases}$$
   Find the mean and variance of the area of the circle.

7. (a) A point X is chosen at random on a line segment AB of length $2a$ with mid-point O. Find the probability that AX, BX and AO can form the sides of a triangle. 4

   (b) 50% of a product have weight 60 kg or more and 10% have weight less than 55 kg. Assuming the distribution of weight to be normal (μ, σ^2), evaluate $(\mu - 3\sigma, \mu + 3\sigma)$. 6
   For a standard normal variate Z, $P(Z \le 1.28) = 0.90$ approximately.

Module-IV

8. (a) Solve the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}, (x > 0, t > 0)$ subject to the conditions 6
   (i) $u(0, t) = 0$ (ii) $u(x, 0) = \begin{cases} 1, & 0 < x < 1 \\ 0, & x \ge 1 \end{cases}$ (iii) $u(x, t)$ is bounded.

   (b) Prove that $xJ_n'(x) = nJ_n(x) - xJ_{n+1}(x)$. 4

9. (a) Obtain the series solution of the differential equation 6
   $\frac{d^2y}{dx^2} + y = 0$, near $x = 0$ such that $y(0) = 1$, $y'(0) = 2$.

   (b) Prove that $\int_{-1}^{1} P_m(x)P_n(x)dx = 0$, if $m \ne n$. 4

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